12.4.2 OPTICS Algorithm

Once given these two definitions, we can easily describe the algorithm; its core idea is similar to DBSCAN, adding points to current cluster only if they are reachable from points already in the cluster. However, OPTICS is also completely different: for starters, it doesn't produce just a flat partitioning of the points, but instead it builds a hierarchical clustering – we'll get back to this in a minute.

OPTICS takes the same parameters as DBSCAN, but ε, in this case, has a different meaning: it's the largest radius that will be considered to determine core and reachability distance. For this reason (and for disambiguation, as we'll see), in the rest of the chapter we will refer to this parameter for OPTICS as `εMAX`. By using this approach, the algorithm simultaneously constructs density-based clusterings for an infinite number of values of ε (all values between `0` and `εMAX`), and hence for an infinite number of densities.

In the formulas in the previous sub-section we have seen that these distances are set to `undefined` if the ε-neighborhood of a point doesn't have enough neighbors; this also means that the reachability distance from a point `q` (to any other core point) will be `undefined` if it's not a core point, with `M-1` neighbors within a radius `εMAX`.

Therefore, if we set `εMAX` to a large value, then we will have less points marked as noise and, in a way, we will leave us more open options, because we will allow larger values for the core distance, and we will have more points with a defined value for reachability distance.